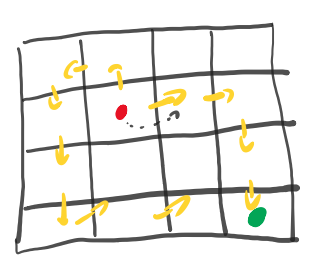


$\{a\} = \{\leftarrow, \downarrow, \rightarrow, \uparrow\}$

• RLHF

• Reward Model.

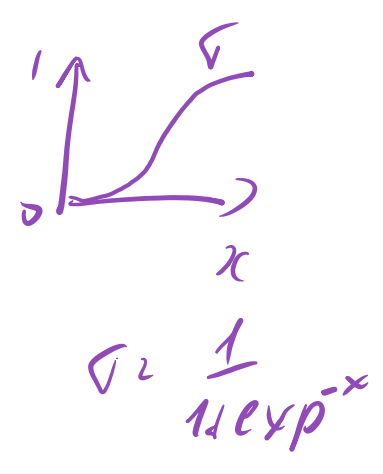
$$\underbrace{\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x,y_w,y_l)\sim D}\left[\log\left(\sigma\left(r_\theta\left(x, y_w\right) - r_\theta\left(x, y_l\right)\right)\right)\right]} \to \max_\theta$$

кол-во генераций

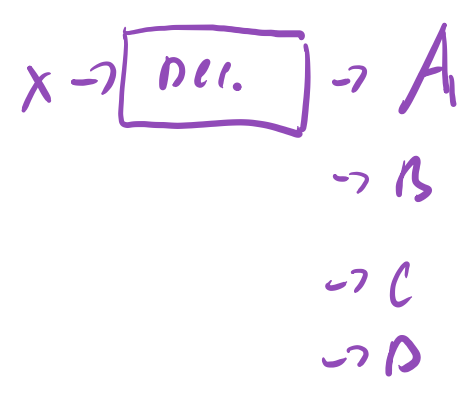
prompt

winner

loser



$\sigma = \frac{1}{1+\exp^{-x}}$



A → B - C - D

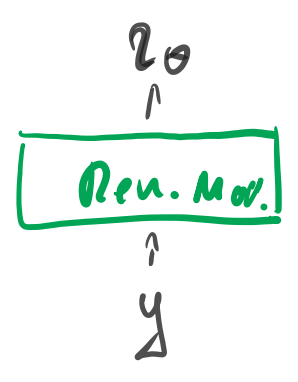
w l

A B

A C

A D

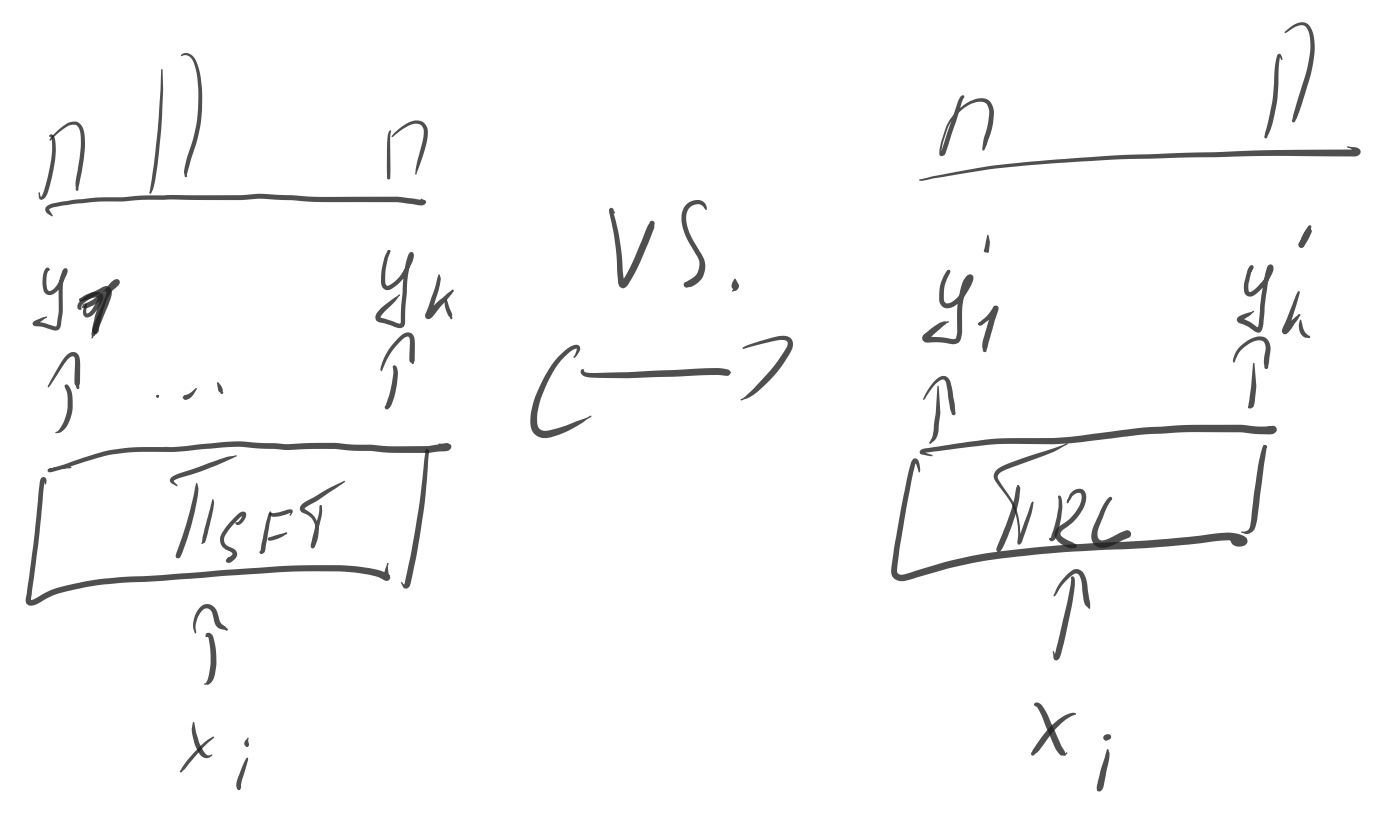
$\binom{K}{2} = C_K^2 = \frac{K(K-1)}{2!}$

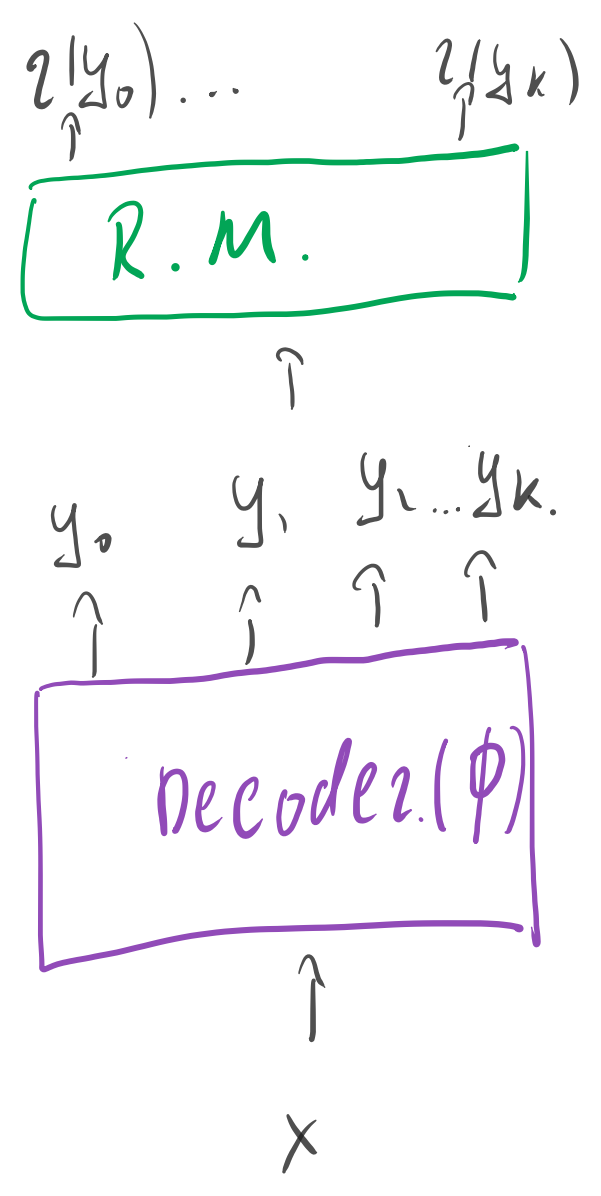


• Proximal policy optimisation

model

$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\text{RL}}}}\left[r_\theta(x,y) - \underbrace{\beta\log\left(\pi_\phi^{\text{RL}}(y\mid x)/\pi^{\text{SFT}}(y\mid x)\right)}_{\text{proximal}}\right]$$





$$J_{x_i} = \sum_{j=1}^{K} \pi_\phi(y_j|x_i)\, r_\theta(x_i, y_j)$$

$$p(y_j) = p(t_1|t_0)\,p(t_2|t_{0,1})\ldots = \prod_{i=0}^{\text{seq-len}} p(t_i|t_{<i})$$

$$J = \sum_{i=1}^{N} J_{x_i} = \sum_{i=1}^{N}\sum_{j=1}^{K} \pi_\phi(y_j|x_i)\, r_\theta(x_i, y_j)$$

$$J_{x_i} = \sum_{j=1}^{K} \pi_\phi(y_j|x_i)\, r_\theta(x_i, y_j) = E_y\, r_\theta(x_i, y) \approx \frac{1}{S}\sum_{l=0}^{S} r_\theta(x, y)$$

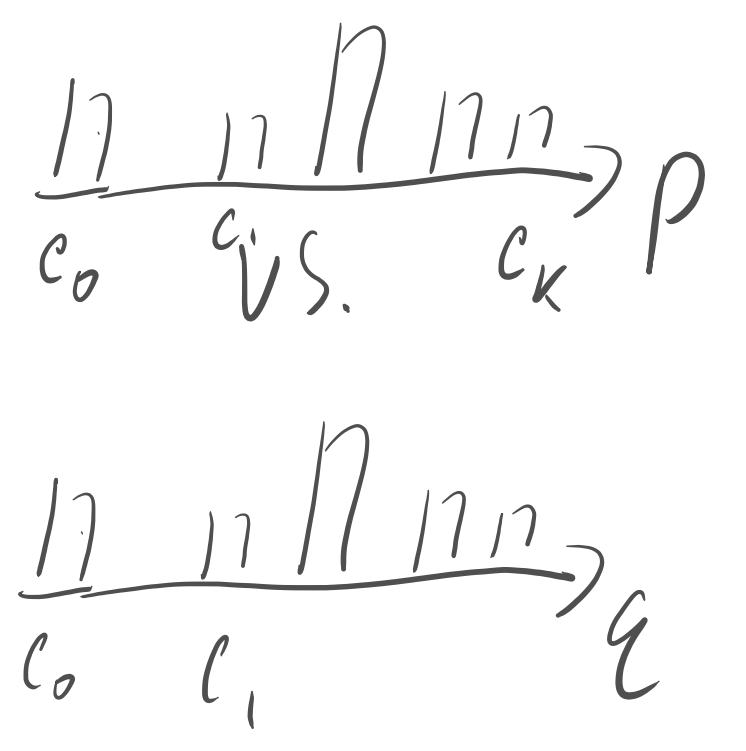
$$\nabla_\phi J = \nabla_\phi \sum_{i=1}^{N} J_{x_i} = \sum_{i=1}^{N} \nabla_\phi J_{x_i} =$$

$$= \sum_{i=1}^{N} \nabla_\phi \frac{1}{S}\sum_{l=0}^{S} r_\theta(x_i, y)$$

$$\nabla_\phi J = \nabla_\phi \sum_{i=1}^{N} J_{x_i} = \sum_{j=1}^{K} \boxed{\nabla_\phi \pi_\phi(y_j|x_i)}\, r_\theta(x_i, y_j) =$$

$$= \sum_{j=1}^{K} \pi_\phi(y_j|x_i)\, \underbrace{\nabla_\phi \log(\pi_\phi(y_j|x_i))\, r_\theta(x_i, y_j)}$$

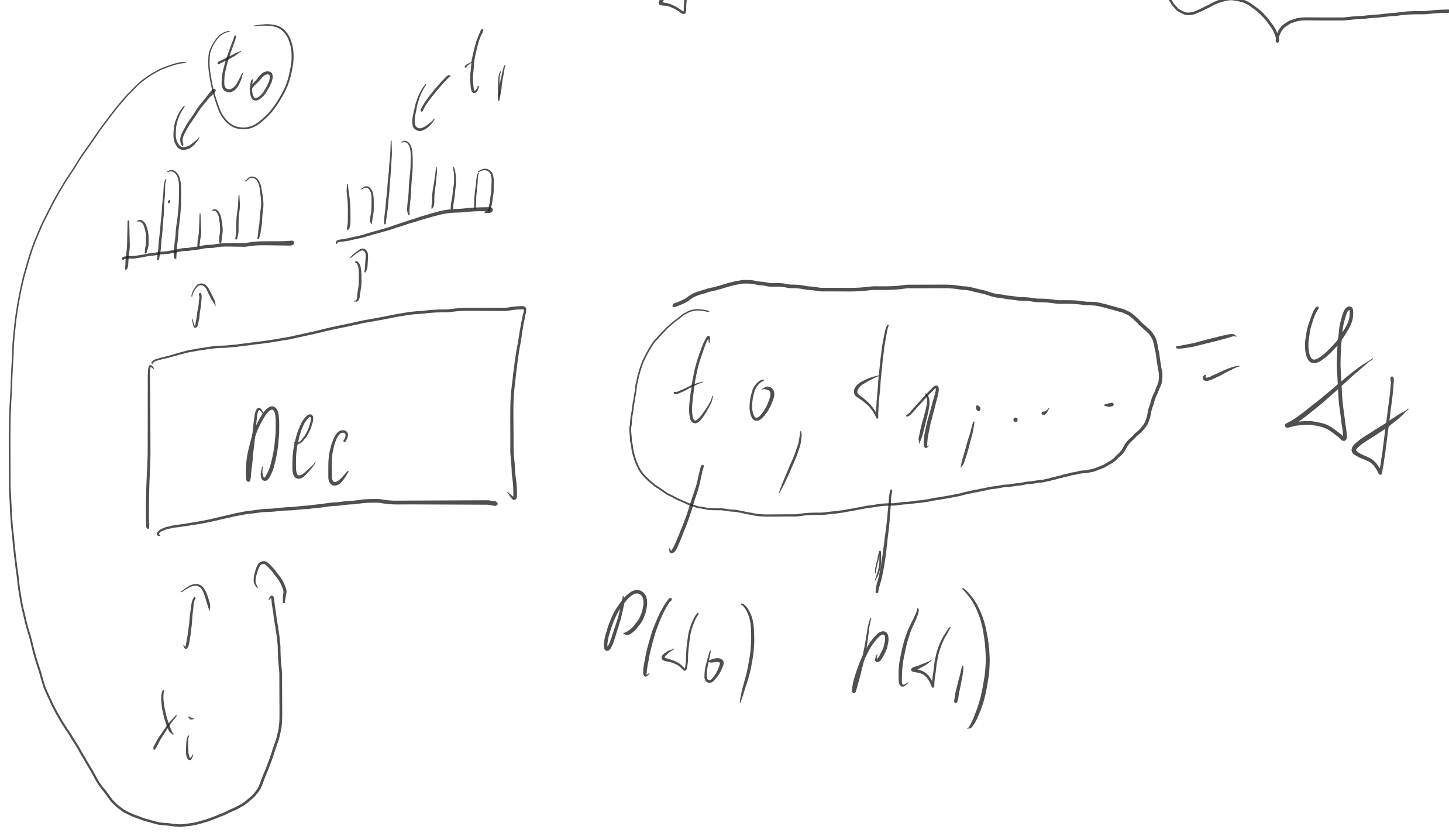
$$KL_{p\|q} = \sum_{i=0}^{K} p(c_i)\log\left(\frac{q(c_i)}{p(c_i)}\right)$$



log-derivative trick.

$$\log'(f(x)) = \frac{1}{f(x)} f'(x) \Rightarrow$$

$$\Rightarrow f'(x) = f(x)\log'(f(x))$$



$$\text{objective}(\phi) = E_{(x,y)\sim D_{\pi_\phi^{\text{RL}}}}\left[r_\theta(x,y) - \beta\log\left(\pi_\phi^{\text{RL}}(y\mid x)/\pi^{\text{SFT}}(y\mid x)\right)\right] + \underbrace{\gamma E_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\text{RL}}(x))\right]}$$